From exponent to risk: power-law tails and the probability of large dengue epidemics in Brazil

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Abstract

Power-law tails are widely reported for dengue and read as a verdict on forecastability: with an exponent near two, extremes are held to be unpredictable. We show that it specifies risk instead, and test it out of sample on weekly data for 5,571 Brazilian municipalities (2010–2026), fitting discrete power laws with Kolmogorov–Smirnov selection of the lower bound under five constructions of the same series (50,391 fits) against an exponential.

Construction determines the verdict: median exponents range from 1.95 to 2.08, and the same counts support a power law in 77% of municipalities pooled over a decade against 23% within single seasons. Epidemic size, the annual total of the episode, retains a genuine tail: the power law is preferred in all 16 national cross-sections (median $\alpha = 2.03$) and in 96% of 353 state-years (median $\alpha = 1.98$), with lower bounds two to three orders of magnitude above the bulk.

Because $\alpha \approx 2$ fixes the exceedance curve, the pair is a two-parameter risk specification: 5.6–16.5% of municipalities exceed ten times the lower bound in a given year, 0.3–2.7% a hundred times it. The exponent is rank-correlated with that bound at 0.89 nationally and 0.64 within states, an association absent in simulation.

Framed as exceedance, prediction is feasible where the panel resolves it: at municipal level, with the label referred to each unit's own history, lagged descriptors, including duration and peak timing, improve the Brier score over calibrated persistence in 4 of 5 origins and are significantly the worse in the fifth, the only contrast satisfying the pre-specified rule, and raise the area under the ROC curve from 0.50 to 0.58, the full ladder reaching 0.65; at state level the same contrast holds in 2 of 5 origins, where the label is pooled across units. The scaling block satisfies the rule at neither level; for size its median differential at $\tau = 0.95$, +4.9%, excludes zero in 2 origins in opposite directions. The tail belongs in a risk curve, not in a predictor.

1 Introduction

The dynamics of infectious disease epidemics have generated an extensive literature, both in mechanistic mathematical modelling and in statistical inference (Siettos and Russo, 2013). The former seeks to represent the causal chain of transmission; the latter seeks to characterise the distribution of observed incidence.

Among the most influential results of the mechanistic tradition is the basic reproduction number R_0, which governs the speed of spread and defines the epidemic threshold; in a vector-borne infection the threshold quantity rests on a generation interval that varies

with temperature, and must be estimated accordingly (Codeço et al., 2018). Because such models are seldom identifiable from incidence alone, their parameters are usually obtained within a calibration framework that returns a distribution rather than a point estimate (Coelho et al., 2011). In the statistical tradition, incidence is most often modelled as a stochastic process, typically Poisson or negative binomial (Becker and Britton, 1999; Li et al., 2012). The central difficulty in either case is overdispersion: surveillance counts generate far more extreme observations than tractable count models admit.

A recurrent claim is that this overdispersion is not merely large but *scale-free*, that incidence follows a power law (Rhodes et al., 1997; Clauset et al., 2009; Neipel et al., 2020). Where transmission is itself power-law-tailed the exponent is not a descriptive curiosity: the survival probability of an outbreak obeys a scaling law whose form is set by that exponent (Falcó and Corral, 2022). The claim matters because a heavy tail implies extremes that most models fail to predict, and because those extremes account for most of the disease burden and most of its economic cost. The quantity that public health planning acts on is the attack ratio, the fraction of the population infected in a season, which has to be inferred from aggregated notification data because serotype-specific immunity is unobserved (Coelho and Carvalho, 2015). Were the claim correct, it would bear directly on forecasting and on the allocation of resources.

The empirical record is inconsistent. Reported exponents for dengue vary widely, some studies find exponential rather than power-law behaviour, and the presence of a heavy tail appears to depend on the study period, on the spatial unit and on the outcome variable. This heterogeneity has generally been attributed to biology or to geography, on the assumption that different places or different years are governed by different regimes. In Brazil the geography has itself moved: dengue has expanded towards higher latitudes and towards municipalities with no prior transmission, so that the set of units entering a cross-section changes from one season to the next (Codeço et al., 2022).

We test a narrower set of claims, and we begin from a different premise: that a power law is not a verdict on forecastability but a *specification of risk*. An exponent of two assigns a determinate probability to every exceedance level, so that the fitted pair $(\alpha, x_{\min})$ summarises the entire risk curve of a municipality using two parameters — in a setting where the observables vary by two orders of magnitude between years. What a heavy tail precludes is that the *magnitude* of the largest event be estimated from the variance; what it supplies is the probability that such an event occurs. These are distinct problems, and a proper scoring rule keeps them distinct.

1.1 Questions addressed

We pose three questions in sequence, each conditional on the answer to the preceding one.

First, *what is the exponent an exponent of?* Any epidemiological series admits several legitimate partitionings — the weekly counts of one municipality, its annual epidemic totals, the set of epidemic sizes across municipalities within one season — and these are different random variables with different supports. We fit the same estimator to the same surveillance data under five constructions and report the exponent together with the diagnostic ($x_{\min}$) that makes it interpretable. Epidemic size, the quantity surveillance services act on, is the construction that survives. Epidemic duration and peak timing are not submitted to that test: both are bounded by the length of the season, so a scale-free law cannot hold for them a priori, and the test would return a verdict about the estimator rather than about the epidemic.

Second, *what does the fitted pair provide?* If $(\alpha, x_{\min})$ is a risk curve, it should reproduce the empirical frequency of large seasons. We test that directly against the same

cross-sections.

Third, *does the tail improve forecasts?* We pose the question in the two forms in which it could plausibly succeed: as a classification of whether a municipality will exceed its own historical upper quartile, which is the risk statement itself; and as a quantile regression of epidemic size. Epidemic duration and peak timing enter not as objects of a scaling claim but as ordinary predictors, available before the season, which may or may not be informative. The evaluation uses a rolling-origin design, a four-rung ablation ladder and a decision rule fixed in advance.

1.2 Contributions

1. **A robust exponent for epidemic size, reported with the diagnostic that makes it interpretable.** Under five constructions of the same surveillance series, the two weekly constructions show what the diagnostic is for: $x_{\min}$ collapses to five cases or fewer in 77% of W-mun and 81% of W-seas fits, so that the power law is estimated on counts of one and two. The size constructions do not collapse, and they agree: the power law is preferred in 16 of 16 national cross-sections and in 96% of 353 state-year cross-sections, with median exponents of 2.03 and 1.98 and median lower bounds of 730 and 169 cases respectively. Reporting $x_{\min}$ beside every exponent is what separates the fits that describe a tail from those that describe the bulk of the data.

2. **The fitted pair as a calibrated risk curve.** Equation (3) maps $(\alpha, x_{\min})$ to $\Pr(S \geq s)$ for every s, and the national fits reproduce the observed exceedance frequency over two decades of epidemic size. This is a decision-relevant output that a point forecast does not provide, and it is obtained from two parameters.

3. **A measured coupling that identifies what the exponent represents.** Within units, α and $\log_{10} x_{\min}$ are rank-correlated at 0.64 (median over 24 states) and at 0.89 across national cross-sections, whereas a simulation with the exponent held fixed recovers no such association. The pair is therefore a low-variance summary of seasonal magnitude: informative about size, but not independent of lagged incidence, which a reading of the exponent as an independent predictor would require.

4. **An out-of-sample answer in the form in which the question can be answered.** The probability of a large season is predictable where the panel is large enough to resolve it: lagged descriptors improve the Brier score over calibrated persistence in 4 of the 5 municipal forecast origins, the only contrast in this study to satisfy the pre-specified rule, whereas at state level the same rung fails on 23 to 25 units per origin. The scaling block adds no incremental skill to either task — 0 of 5 origins for size, 1 of 5 for municipal exceedance — and leaves the upper quantile of the size forecast where it was. The scaling description is thus justified as a risk statement and not as a predictor.

2 Methods

2.1 Data sources and study period

We analyse weekly dengue case counts assembled by the InfoDengue/Mosqlimate platform (Araujo et al., 2025) for the period from 1 January 2010 to 22 January 2026, for every

Brazilian municipality represented in the platform. The analysed variable is the nowcast-adjusted estimated number of cases, that is, the platform's reconstruction of the number of infections occurring in each week after correcting for reporting delay by a hierarchical model of the delay distribution (Bastos et al., 2019); it is not the raw notification count. The series were projected onto a common epidemiological week ending on Sunday. Weeks without a reported case are retained in the series but excluded from distributional fits, since the power-law family is defined on positive support.

Descriptors of each epidemic season were obtained from the platform's seasonal decomposition, which partitions a municipal series into epidemic episodes and reports, for each episode, its total number of cases, its duration in weeks and the week in which it peaked. These descriptors are available for 16 years, 2011 to 2026, the last of them bounded by the data cut-off of 22 January 2026; the fit pooled over all years is reported separately and is excluded from every year count, so that each national cross-section corresponds to one calendar season. Only epidemic size is fitted as a power law; duration and peak timing enter the analysis as predictors of size.

Municipal population figures are the estimates distributed with the platform (5,571 municipalities, one record with a zero entry excluded). Population enters no power-law fit; it is used to compute the descriptive correlations of Section 3.3 and, as $\log_{10}$ population, as one predictor in the forecasting models of Section 2.8.

2.2 Definitions and notation

Several quantities recur throughout the paper, and their meaning is fixed here once and for all. Table 1 collects the notation.

Weekly count. Let $c_{i,w}$ denote the estimated number of dengue cases in municipality i during epidemiological week w. It is the primitive observation on which every construction in this paper is built.

Season. A season, or epidemic year, t is the set of epidemiological weeks assigned to calendar year t; it contains at most 53 weeks. A season is indexed by the calendar year of its weeks, not by the year in which an epidemic peaks.

Epidemic size. Let $S_{i,t}$ denote the total number of cases in the epidemic episode of municipality i in season t, as identified by the seasonal decomposition. We refer to $S_{i,t}$ as the *epidemic size*, and we use *annual total* as a synonym for it and for no other quantity. In particular, $S_{i,t}$ is neither the weekly count $c_{i,w}$ nor the calendar-year sum of $c_{i,w}$, which would also include cases notified outside any identified episode. The distinction is operational as well as definitional: an episode is identified only where the decomposition detects one, so municipality-seasons without a detected episode contribute no observation to the size distribution, whereas the weekly counts exist for every municipality and week.

Cross-section. A cross-section at season t is the set $\{S_{i,t}\}$ of epidemic sizes of the municipalities within a stated spatial scope. The *national cross-section* contains every Brazilian municipality with an identified episode in t; a *state-year cross-section* contains the municipalities of one state in one season. A cross-section is therefore a sample of units at a fixed time, in contrast to a series, which is a sample of times at a fixed unit.

Construction. A construction is the triple (fitted variable, sampling unit, dimension of pooling) that determines which random variable is estimated and what each observation of the fitted sample represents.

Lower bound and bulk. The lower bound $x_{\min}$ is the threshold above which the power-law form is assumed to hold, and the *bulk* is the part of the empirical distribution below it. A fit is *tail-informative* when $x_{\min}$ lies above the bulk, in the operational sense of Section 2.6.

Duration and peak timing. $D_{i,t}$ denotes the duration of the episode in weeks and $P_{i,t}$ the epidemiological week of its maximum.

Table 1: Notation. The first group defines the observed quantities, the second the parameters of the fitted law, and the third the design of the predictive evaluation.

Symbol	Meaning
$c_{i,w}$	estimated cases in municipality i, epidemiological week w
t	season, indexed by calendar year
$S_{i,t}$	epidemic size: total cases of the episode of i in t
$D_{i,t}$, $P_{i,t}$	duration (weeks) and peak week of that episode
$R_{0,t}$	effective reproduction number reported for season t
α	power-law exponent
$x_{\min}$	lower bound of the power-law regime
R	normalised log-likelihood ratio, power law against exponential
τ	quantile level of the forecast
M0–M3	rungs of the ablation ladder

2.3 Five aggregation levels

The unit of analysis is the sample of positive observations passed to the estimator. Table 2 defines the five aggregation levels used; each is a construction in the sense of Section 2.2, and the two terms are used interchangeably. Two variables are fitted, both derived from the same municipal series: the weekly count $c_{i,w}$ at constructions W-mun and W-seas, and the epidemic size $S_{i,t}$ at constructions S-mun, S-state and S-nat. Within each group the constructions differ only in the dimension along which the variable is pooled, which is what allows a contrast between two of them to be read as the effect of that dimension alone.

Every handle is read as a variable and a scope. The initial letter names the fitted variable — W for the weekly count, S for the epidemic size — and the remainder names the scope over which observations are pooled: *mun* for a single municipality, *seas* for a single season of a single municipality, *state* for the municipalities of one state in one season, and *nat* for the municipalities of the country in one season. The handles name constructions and are not random variables: only $c_{i,w}$ and $S_{i,t}$ enter the likelihood. They are used throughout for the five constructions and also label the panels of Figures 1, 2 and 3.

A construction is fixed by two independent decisions. The first selects the variable that enters the likelihood: the weekly count $c_{i,w}$ as observed, or the season total $S_{i,t}$, obtained by summing the weeks of the epidemic episode delimited by the seasonal decomposition described in Section 2.1. The second selects the sample over which that variable is pooled: the weeks of one municipality, taken over the complete series or over a single season; the seasons of one municipality; or the municipalities of one state or of the whole country within a single season. The five constructions of Table 2 are the combinations in use, not the full crossing of the two decisions; a pairing omitted from the table, such as the weekly counts of a national cross-section, is one that no contrast in this paper requires.

Constructions W-mun and W-seas differ only in the temporal window over which the weekly counts are pooled: W-mun uses the complete available series of a municipality (January 2010 to January 2026), whereas W-seas restricts the sample to a single season,

Table 2: The five aggregation levels. "Fitted variable" is the random variable entering the estimator; "sampling unit" is what each observation of the fitted sample represents; "pooling" is the dimension along which observations are collected. W-mun and W-seas fit the weekly count over different windows; S-mun, S-state and S-nat fit the epidemic size over different dimensions.

Handle	Fitted variable	Sampling unit	Pooling	Fits
W-mun	weekly count c	municipality	weeks, full series	5,264
W-seas	weekly count c	municipality-season	weeks within one season	43,673
S-mun	epidemic size S	municipality	seasons, ten or more	1,085
S-state	epidemic size S	municipality in state-year	municipalities of a state	353
S-nat	epidemic size S	municipality in national-year	municipalities nationally	16

of at most 53 weeks. Their sample sizes differ by roughly an order of magnitude, which is why the comparison between them is made on shares rather than on counts.

Constructions S-mun, S-state and S-nat fit the same variable, $S_{i,t}$, and differ only in the dimension of pooling. S-mun pools the seasons of one municipality and is therefore a statement about how a single municipality's epidemic sizes vary over time; a municipality enters this construction only if at least ten seasons are available. S-state and S-nat pool municipalities within a period, at state and national scope respectively, and are therefore statements about how sizes vary across space in a given season. This nesting is deliberate: each contrast reported in Section 3 is a single step within one of the two groups, so that a difference in the inferred regime is attributable to the construction rather than to a change of data source, period or variable.

2.4 Discrete power-law estimation

For each sample we fit a discrete power law by maximum likelihood following Clauset et al. (2009). The fitted probability mass function for integer $x \geq x_{\min}$ is

$$p(x) = \frac{x^{-\alpha}}{\zeta(\alpha, x_{\min})}, \tag{1}$$

where $\zeta(\alpha, x_{\min})$ is the generalised (Hurwitz) zeta function, which normalises the discrete mass. The exponent α is estimated by numerical maximisation of the likelihood, and the lower bound $x_{\min}$ is selected jointly by minimising the Kolmogorov–Smirnov distance between the empirical and fitted cumulative distributions over all candidate values of $x_{\min}$. Because the observations are counts, the discrete formulation is used throughout; the continuous approximation is not employed at any construction.

One property of the estimator must be stated for the table of results to be read correctly. The numerical optimisation constrains α to the interval $(1, 3]$: an estimate cannot exceed 3, and a sample whose likelihood is maximised beyond that value returns $\alpha = 3$ rather than its unconstrained maximum. The bound is immaterial for epidemic size, the variable on which the substantive conclusions rest. At the size constructions no fit approaches it, the largest values being 2.94 at state level and 2.25 at national level; at the two weekly constructions it is reached by 12 of the 48,937 fits, or 0.02%. The exponents reported in this paper are accordingly unconstrained maxima and not boundary values.

The choice of $x_{\min}$ is not a nuisance parameter. It determines which fraction of the data the model actually describes, and therefore whether the fitted exponent refers to a

tail at all. We therefore retain $x_{\min}$ for every fit and report its median and the proportion of fits with $x_{\min} \leq 5$ at each construction.

2.5 Comparison against an alternative distribution

A power law can always be fitted; the question is whether it describes the data better than a plausible alternative. We compare each power-law fit against an exponential distribution, which is the natural null for epidemiological counts because it corresponds to a constant hazard of termination and generates a light tail. The comparison follows the normalised likelihood-ratio procedure of Clauset et al. (2009),

$$R = \sum_{i=1}^{n} \left[\ln p_{\mathrm{PL}}(x_i) - \ln p_{\exp}(x_i) \right], \tag{2}$$

normalised so that its magnitude is comparable across samples of different size. The sign of R indicates the preferred model, with $R > 0$ favouring the power law and $R < 0$ the exponential, and its associated p-value quantifies whether the sign is distinguishable from sampling noise.

We adopt a deliberately conservative reporting convention. A construction is described as *power-law supported* only when $R > 0$ and $p < 0.05$, as *exponential supported* only when $R < 0$ and $p < 0.05$, and as *inconclusive* otherwise. This prevents the common error of counting a positive but non-significant R as evidence for scaling.

2.6 Identification diagnostic

A fit is flagged as *tail-informative* when $x_{\min}$ exceeds the bulk of the observed distribution, operationalised as $x_{\min} > 5$ cases for weekly counts and as $x_{\min}$ above the median observed value for the size constructions. Where $x_{\min} \leq 5$, the fitted power law is being asked to describe the entire empirical distribution, including the small-count regime in which the discrete power law is not a plausible generative model, since counts of one or two cases arise from reporting mechanics rather than from scale-free branching. Such fits are retained in the analysis but excluded from substantive interpretation, and the proportion concerned is reported explicitly.

2.7 From description to prediction: the exceedance curve

A fitted power law is not merely a description of a histogram; it is a statement about exceedance probabilities. For integer $s \geq x_{\min}$,

$$\Pr(S \geq s) \;=\; \frac{\zeta(\alpha, s)}{\zeta(\alpha, x_{\min})}, \tag{3}$$

so that a single pair $(\alpha, x_{\min})$ parameterises the probability that a municipality experiences an epidemic of at least any given size. Four consequences of Equation (3) shape the design of the forecasting evaluation.

First, the exponent summarises *risk* rather than *activity*. Two municipalities with the same expected annual incidence but different α face different probabilities of a catastrophic season, and that difference is invisible to the mean. The fitted models quantify the claim rather than merely asserting it: applied to the 16 national size fits, Equation (3) implies that between 5.6% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given year, and that between 0.3% and 2.7% exceed one hundred times $x_{\min}$. These are falsifiable

implications of the description, and Figure 8 confronts them with the empirical exceedance frequencies.

Second, α is a low-variance quantity *relative to its target*. Across the national cross-sections it takes values between 1.78 and 2.25, and it is essentially unrelated to municipality size ($r = -0.043$). Annual case counts over the same period vary by two orders of magnitude. The dispersion of $\hat{\alpha}$ is not zero — it exceeds sampling noise by roughly a factor of five in standard deviation, and it co-varies with the selected $x_{\min}$ (Section 3.3) — but it is small along precisely the dimension in which the target is volatile. That is the specific reason to carry α into the predictive models rather than treating it as a descriptive curiosity, and the same coupling is the reason the ablation, rather than the correlation, has to decide the question.

Third, $\alpha \approx 2$ has a direct methodological consequence for evaluation. The second moment of a power law with $\alpha \leq 2$ does not exist in the continuous limit, and at $\alpha \approx 2$ the sample mean is unstable, so that squared-error losses are decided by the few largest outbreaks. A model that is adequate everywhere except in the tail can therefore appear excellent or inadequate depending on which years fall in the test set. We consequently evaluate on quantile (pinball) losses and on the classification of tail events, and treat the root mean squared error as a secondary, tail-dominated diagnostic.

Finally, the predictive hypothesis is stated in a form that can fail. If $(\alpha, x_{\min})$ carries information about the coming season beyond what lagged counts already encode, a model containing them should outperform an otherwise identical model without them, out of sample. If instead the exponent is close to a deterministic function of the lower moments of the same series, the two models should be indistinguishable, and the descriptive result would carry no forecasting content. Section 2.11 specifies the comparison that decides between these outcomes, and Section 2.12 lists the four ways in which such a comparison could be falsified and how each is prevented.

2.8 Predictive target and predictors

The forecasting problem is posed at the level of an administrative unit, a municipality or a state, and a target season t: every predictor is measured in season $t-1$ or earlier, and the target is a property of the season of year t. There is one target, the epidemic size S_t defined in Section 2.2, because it is the quantity surveillance services act on and the only one of the three seasonal descriptors with unbounded support. Epidemic duration D_t and peak timing P_t are predictors, not outcomes: they describe the phenology of the previous season, and the ladder asks whether that phenology helps forecast the magnitude of the next one.

Predictors fall into six blocks, listed in Table 3: lagged seasonal descriptors of the previous season, which include its duration and peak timing; the scaling descriptors $(\alpha_{S,t-1}, x_{\min,S,t-1})$ obtained from the power-law fit to the previous season; the historical trend $\dot{\alpha}_{t-1}$, defined as the ordinary least-squares slope of α_S over all seasons strictly before t and set to zero where fewer than two such seasons exist; the epidemiological driver $R_{0,t-1}$ reported by the seasonal decomposition; $\log_{10}$ population; and eight quarterly climate aggregates from year $t-1$, namely mean minimum temperature and mean minimum humidity in each of the four quarters. Units lacking the mandatory lags, that is without a previous season, are excluded rather than imputed; the optional $\dot{\alpha}$ is set to zero where it is undefined.

Table 3: Predictor blocks. All predictors refer to season $t-1$ or earlier. The scaling block is the object of the ablation of Section 2.11.

Block	Variables	Derivation
Lagged descriptors	S_{t-1}, D_{t-1}, P_{t-1}	seasonal decomposition
Scaling descriptors	$\alpha_{S,t-1}$, $x_{\min,S,t-1}$	power-law fit to season $t-1$
Scaling trend	$\dot{\alpha}_{t-1}$	least-squares slope of α_S before t
Epidemiological	$R_{0,t-1}$	seasonal decomposition
Demographic	$\log_{10}(\text{pop})$	municipal population estimate
Climate	$\bar{T}_{Q1..Q4}$, $\bar{H}_{Q1..Q4}$	mean minimum temperature and humidity

2.9 Regression estimator

The choice of estimator is a design decision in this problem and not a tuning detail, because the quantity of interest can lie outside the range of the data to which the model is fitted. Under $\alpha \approx 2$ the season of greatest public-health interest is systematically larger than any season in the training record, so the forecast must be allowed to take a value beyond the largest response observed. The primary estimator is therefore a quantile regression on a cubic spline basis with linear boundary extrapolation (Koenker and Bassett, 1978). It minimises the pinball loss. Outside the observed range the boundary segment continues linearly, which on the $\log(1+S)$ scale is a multiplicative extrapolation on the scale of cases. The two predictors expressed in cases, S_{t-1} and $x_{\min,t-1}$, are logged before entering the basis, without which the extrapolation would be linear in the wrong scale. The L_1 penalty is 10^{-4}, deliberately weak.

A random forest of 150 bootstrapped trees (Breiman, 2001), fitted to the same predictors, is reported alongside the primary model in Figure 11. All quantile models are fitted on the grid $\tau \in \{0.05, 0.10, \ldots, 0.95\}$, which yields a full predictive distribution per unit and season rather than a point forecast, and all stochastic components were seeded identically across runs.

2.10 Classification of large seasons

The second predictive task is posed as a decision problem rather than as an estimation problem: will unit i experience, in season t, a season larger than the p-th percentile of its own historical distribution? The target is

$$Y_{i,t} \;=\; \mathbb{I}\big[\, S_{i,t} > Q_p\big(\{S_{i,k}\}_{k<t}\big) \,\big]\,, \tag{4}$$

with $p = 0.75$ throughout. The threshold is a within-fold quantity: it is recomputed from the training seasons of each fold and of each split, never from the full sample, so that the label of a test observation cannot depend on information unavailable to the model. The same discipline governs the class-imbalance weight, set to the ratio of negative to positive observations in the training fold.

The scope over which the percentile is taken is set by what each level can support. At municipal level Q_p is computed from each unit's own training seasons, and 3,976 of the 5,571 municipalities supply the five seasons that estimate requires. At state level the panel supplies 23 to 25 units per origin, too few for a within-unit percentile, and Q_p is taken across the states of the fold's training seasons. The two labels therefore answer slightly different questions, and the state-level task is the closer of the two to a ranking by unit size (Section 3.5).

The classifier is a penalised logistic regression on the same spline basis, with the class imbalance handled by weighting the training rows by the negative-to-positive ratio. As with the regressor, the specification must remain informative beyond the range covered by the training rows, and the logistic form does so by extrapolating linearly in log-odds, so that a season far outside the range of the panel is not forced towards the base rate observed in training. Validation is an expanding-window split defined on the season index: fold j trains on every unit-season earlier than v_j and validates on season v_j alone, where the v_j are the $k = \min\{5,\, n_{\text{seasons}} - 2\}$ most recent training seasons. A fold therefore exists only once at least two earlier seasons are available, and the number of folds is capped at five. Because the folds are a function of the season index rather than of arbitrary ordering, they are invariant to the ordering of the observations.

This distinction is not cosmetic, and it is the step of the design whose failure is silent. A split defined by arbitrary ordering rather than by the season index produces, in a frame ordered by unit, a grouping by municipality: each fold then trains on some municipalities across the whole calendar and validates on others over the same calendar, so that validation contains no season absent from training. The quantity estimated becomes transportability to new municipalities rather than temporal generalisation, and the estimate depends on an undeclared ordering of the rows. The defect is narrower but not absent when the frame happens to be ordered by season, since contiguous blocks then still cut through calendar years. The split is therefore indexed explicitly by the season, and any validation season that is not strictly later than the training seasons of its fold is treated as an error rather than as a warning. The decision threshold applied to the predicted probability is chosen by Youden's J statistic (Youden, 1950), computed on the out-of-fold predictions rather than on the training fit, and then held fixed for the test set. We report the area under the ROC curve, the area under the precision–recall curve, which is the more informative summary under class imbalance, the Brier score (Gneiting and Raftery, 2007), the F_1 score at the Youden threshold and a calibration curve, and we test differences in the area under the curve between nested predictor sets by the method of DeLong et al. (1988).

Feature attribution is measured by held-out permutation importance on the Brier score, reported both per predictor and for the scaling block permuted as a unit (Breiman, 2001): under the collinearity documented in Section 3.3, a per-column ranking alone understates the importance of a block whose members can substitute for one another.

2.11 Evaluation protocol and ablation ladder

A single temporal split cannot distinguish a predictor that helps from a predictor that happened to help once. We therefore adopt a rolling-origin design: for each forecast origin $y \in \{2021, \ldots, 2025\}$ the model is trained on all observations with season earlier than y and evaluated on season y, and the per-origin losses are pooled. This provides five out-of-sample seasons and allows skill to be read as a function of the history available to the model.

Within each origin, the incremental value of the scaling description is isolated by the ablation ladder of Table 4. Each rung adds one block of predictors to the preceding one, so that a contrast between consecutive rungs is attributable to the block that was added. The rung of substantive interest is M3 against M2, which differ *only* in the presence of $(\alpha_{S,t-1}, x_{\min,S,t-1}, \dot{\alpha}_{t-1})$.

Forecast values at M0 are scaled by the empirical τ-quantile of the within-unit season-on-season ratio, estimated on the training seasons, so that the same benchmark can be evaluated at every quantile level rather than only at the median. At $\tau = 0.5$ the scale

Table 4: The ablation ladder. M0 is the benchmark that any claim of skill must beat; successive rungs add one predictor block, so that consecutive contrasts are attributable to that block. The step from M2 to M3 isolates the contribution of the power-law scaling descriptors.

Rung	Name	Predictors
M0	Persistence	$\hat{S}_t = S_{t-1}$; training base rate for the classifier
M1	Lagged descriptors	M0 predictors + D_{t-1}, P_{t-1}, $R_{0,t-1}$, $\log_{10}(\text{pop})$
M2	+ climate	M1 + quarterly $\bar{T}$, $\bar{H}$ from season $t-1$
M3	+ scaling	M2 + $\alpha_{S,t-1}$, $x_{\min,S,t-1}$, $\dot{\alpha}_{t-1}$

factor is close to unity and M0 reduces to plain persistence; at the other levels this is the honest analogue of persistence and not a deliberately weak comparator.

M0 is deliberately naive, because a benchmark that is already sophisticated manufactures skill for whatever is added to it. Skill is reported as

$$\text{SS} \;=\; 1 - \frac{L(\text{model})}{L(\text{M0})}, \tag{5}$$

on the primary loss of each task: the pinball loss at $\tau = 0.5$ for the regression target and the Brier score for the classifier, both of which are proper scoring rules (Gneiting and Raftery, 2007). Secondary losses are the mean absolute error and the root mean squared error of the point forecasts, the mean absolute scaled error of Hyndman and Koehler (2006) to make errors comparable across units of very different size, and the pinball loss at each τ of the grid.

Because units differ by orders of magnitude in scale, all model-versus-model comparisons are made on paired loss differentials computed within the same unit and season; their uncertainty is estimated by a bootstrap that resamples *units* rather than observations, preserving the within-unit correlation across seasons, with 2,000 replicates. For the comparison pooled across origins we additionally report the Diebold–Mariano statistic (Diebold and Mariano, 1995) with a Newey–West correction, since the loss series are serially correlated by construction.

The decision rule is fixed in advance. The scaling block is judged to add predictive value only if M3 improves on M2 on the primary loss with a 95% bootstrap interval for the paired difference that excludes zero, in the same direction, for at least 3 of the 5 forecast origins. This is stricter than a single significant pooled test, and deliberately so: a pooled test can be carried by one anomalous season, which is precisely the failure mode that a heavy-tailed target invites.

2.12 Leakage controls

Four sources of leakage are structurally possible in this design, and each is closed explicitly.

1. **Contemporaneous scaling descriptors.** $\alpha_{S,t}$ is estimated from the weekly counts of season t, which are part of the target. Only $\alpha_{S,t-1}$ enters as a predictor, and $\dot{\alpha}_{t-1}$ is computed exclusively from seasons strictly before t.

2. **The classification threshold.** Q_p is computed inside each fold from training seasons only. Computing it once on the full sample would leak the distribution of the test seasons into the labels of the training seasons, inflating apparent skill by construction.

3. **Cross-sectional information.** The cross-section of municipalities within season t is an outcome of that season and is therefore never a predictor; the scaling descriptors always come from the previous season's cross-section.

4. **The validation split itself.** Folds are defined on the season index (Section 2.10), so that no fold trains on a season at or after the season it validates on. This is the only one of the four channels that fails silently in a plausible implementation — every number still resembles a respectable validation score — and it is therefore enforced rather than merely documented.

2.13 Data availability

The municipal weekly series and the derived seasonal descriptors are those distributed by the InfoDengue/Mosqlimate platform (Araujo et al., 2025). The fitted parameters and the evaluation outputs underlying every table and figure reported here are available from the corresponding author on request.

3 Results

Table 5 summarises the 50,391 fits across the five constructions. We read it in the order of the questions posed in Section 1.1 and answered below: what the exponent is an exponent of, what the fitted pair provides as a risk statement, and what it adds to a forecast. Figure 1 shows the estimator's output at each construction, Figure 2 the corresponding verdicts, and Figure 3 the diagnostic that separates the constructions worth interpreting from those that are not.

3.1 The construction determines the verdict

One estimator, one body of surveillance data, five constructions, and verdicts that range from unanimous support for a power law (S-nat, 16 of 16 seasons) to the 23% of fits favouring a power law at W-seas. Median exponents span 1.95 to 2.08; individual estimates of the lower bound range from one case to 4×10^4, four orders of magnitude. No two analyses in the table differ in data source, period or variable: W-mun and W-seas are the same variable over different windows, and S-state and S-nat the same variable at different spatial scopes.

Table 5: Power-law fits under five constructions of the same municipal series. α median with interquartile range; $x_{\min}$ median; $\mathrm{PL}_{\mathrm{pref}}$ = proportion with $R > 0$; $\mathrm{PL}_{\mathrm{sig}}$ = proportion with $R > 0$ and $p < 0.05$; $\mathrm{Exp}_{\mathrm{sig}}$ = proportion with $R < 0$ and $p < 0.05$. Shares are percentages. The two weekly constructions produce apparent support that the $x_{\min}$ diagnostic removes; the cross-sectional constructions S-state and S-nat are the ones where the support survives both tests, and they are the object of the rest of the paper.

Construction	n	α	IQR	$x_{\min}$	$\mathrm{PL}_{\mathrm{pref}}$	$\mathrm{PL}_{\mathrm{sig}}$	$\mathrm{Exp}_{\mathrm{sig}}$
W-mun, weekly, city	5,264	1.95	1.75–2.29	2	77	47	3
W-seas, weekly, city-year	43,673	2.07	1.73–2.46	2	23	3	22
S-mun, size, city	1,085	2.08	1.81–2.49	300	52	4	3
S-state, size, state-year	353	1.98	1.80–2.15	169	96	52	0
S-nat, size, national-year	16	2.03	1.90–2.10	730	100	100	0

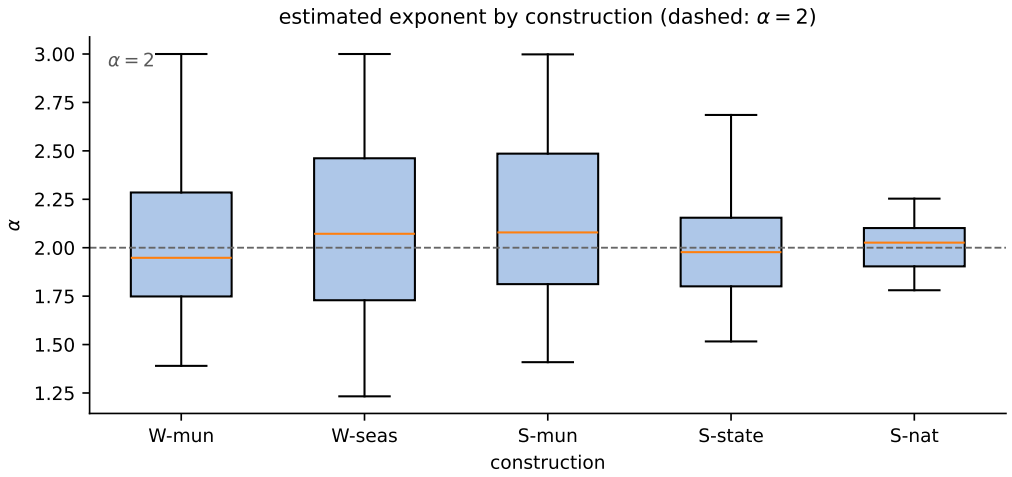


Figure 1: Distribution of the estimated exponent α at each construction. Boxes give the median and interquartile range across fits, whiskers extend to 1.5 times the interquartile range, and the dashed line marks $\alpha = 2$. The two weekly constructions and the three size constructions each fit a single variable, so that the movement of the box within a group is the movement of the construction and not of the object. Sizes concentrate near $\alpha \approx 2$ throughout, with the weekly constructions producing the widest dispersion because most of their fits describe the bulk rather than a tail.

Two features disqualify the weekly constructions as a basis for a scaling claim, and both are visible without a further experiment. Where the estimated lower bound falls within the bulk, the fitted law describes the whole distribution and the language of tails is vacuous: 77% of W-mun and 81% of W-seas fits have $x_{\min} \leq 5$, so that the model is estimated on counts of one and two cases. Second, the verdict changes with the window: pooling the same weekly counts over a decade instead of a single season reduces the proportion of fits favouring a power law from 77% to 23% and raises the proportion significantly favouring the exponential from 3% to 22%. No transmission parameter differs between those two numbers (Figure 5). The corresponding consequence for individual fits is shown in Figure 4.

3.2 Epidemic size follows a power law with an exponent near two

The construction that survives both criteria is the one that matters operationally: the distribution of epidemic size across units within a season. At the national cross-section, all 16 annual fits favour the power law and all are significant, with median $\alpha = 2.03$ (IQR 1.90–2.10) and median $x_{\min} = 730$ cases. The exponent is stable, ranging from 1.78 to 2.25 over sixteen seasons in a period in which annual case counts vary by two orders of magnitude. At state level the picture is consistent and weaker in significance, as expected from smaller cross-sections: the power law is preferred in 96% of the 353 state-year cross-sections and significant in 52%, with median $\alpha = 1.98$ and median $x_{\min} = 169$ cases.

This is the object with which the remainder of the paper works, and the reason is worth stating precisely. The median lower bound of 730 cases at national level and 169 at state level places the fit well above the bulk of a distribution whose observations span four

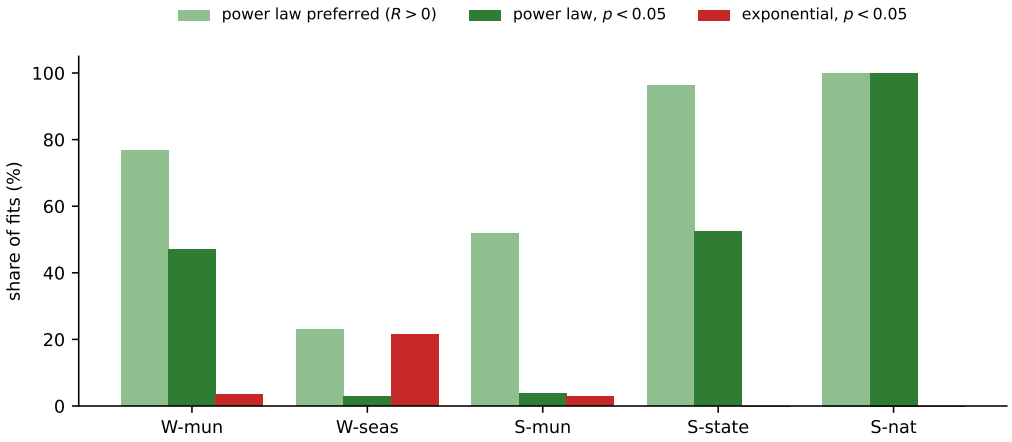


Figure 2: Model comparison by construction. For each construction, the proportion of fits in which the power law is preferred ($R > 0$), the proportion in which it is preferred *and* significant at $p < 0.05$, and the proportion in which the exponential is significant ($R < 0$ and $p < 0.05$). Constructions are ordered from those on individual series (left) to the cross-sectional ones (right). The two weekly constructions produce apparent power-law support that the $x_{\min}$ diagnostic removes; the size constructions are the only ones whose support survives both criteria.

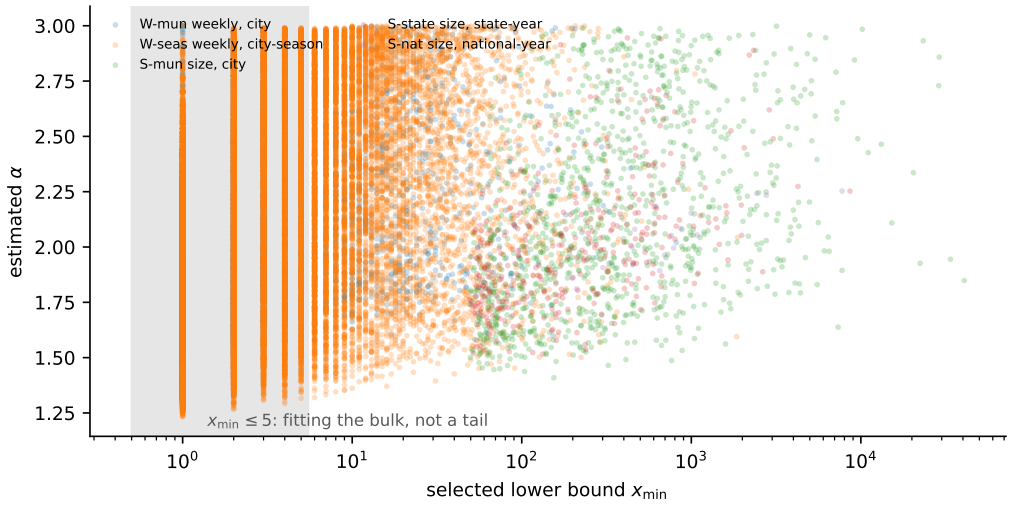


Figure 3: Identification diagnostic. Estimated exponent α against the selected lower bound $x_{\min}$ on a logarithmic scale, one point per fit, coloured by construction. The shaded band marks $x_{\min} \leq 5$, where the fitted power law is asked to describe the bulk of the distribution rather than a tail. The two weekly constructions lie almost entirely inside that band, whereas the size constructions lie two to three orders of magnitude to its right.

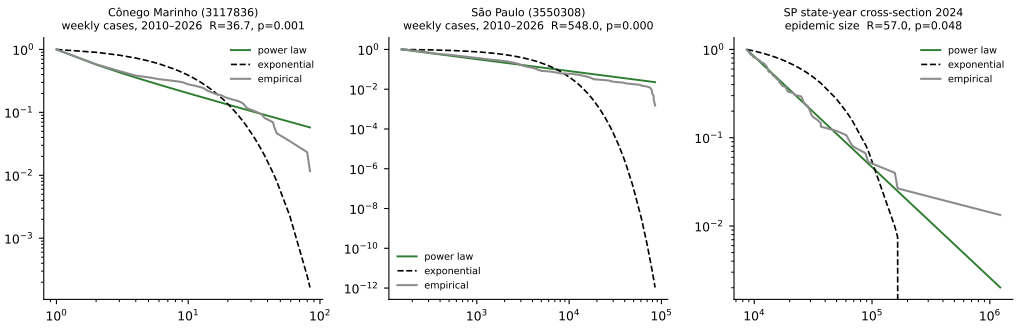


Figure 4: Individual fits at three constructions, each with the power law (solid) and the exponential alternative (dashed) superimposed. (a) Weekly counts of a small municipality, where the selected $x_{\min}$ falls to 2 and the power law is fitted to the entire distribution; (b) weekly counts of a large state capital; (c) a state-year cross-section of epidemic sizes, where $x_{\min}$ lies in the hundreds of cases. All three are produced by the same estimator applied to different objects, and only the third describes a tail.

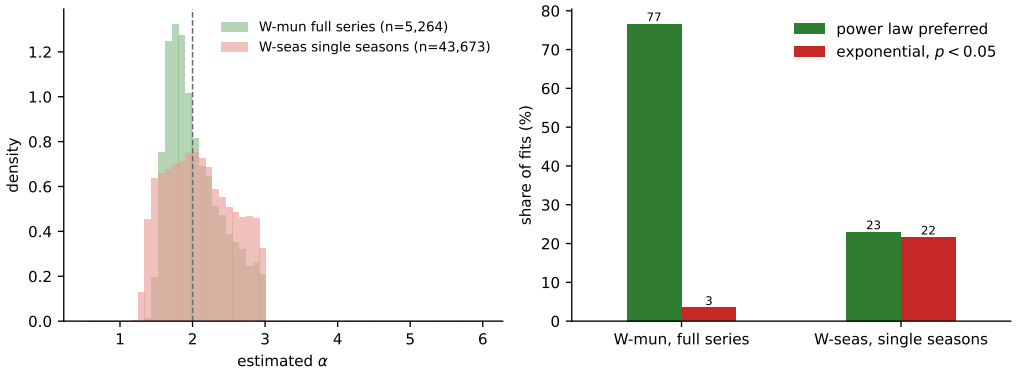


Figure 5: The window reverses the verdict on identical data. Left, density of the estimated exponent from the full municipal series (W-mun) and from the same series partitioned into single seasons (W-seas). Right, the corresponding verdicts: the share of fits favouring the power law falls from 77% to 23%, while the share significantly favouring the exponential rises from 3% to 22%.

orders of magnitude: what is described is the upper region in which the risk is located, not the body of small seasons. This is also the sense in which the two groups of constructions are not interchangeable. At $x_{\min} = 2$ the fitted law describes counts of one and two cases; at $x_{\min} = 730$ it describes the upper decade of a distribution. Two analyses reported as agreeing on the exponent can therefore be describing entirely different objects: W-mun and S-mun have median exponents of 1.95 and 2.08, a difference of 0.13, while their median lower bounds differ by a factor of 150. The empirical complementary cumulative distribution of Figure 6 shows the same fact graphically, as a straight line over two decades above $x_{\min}$.

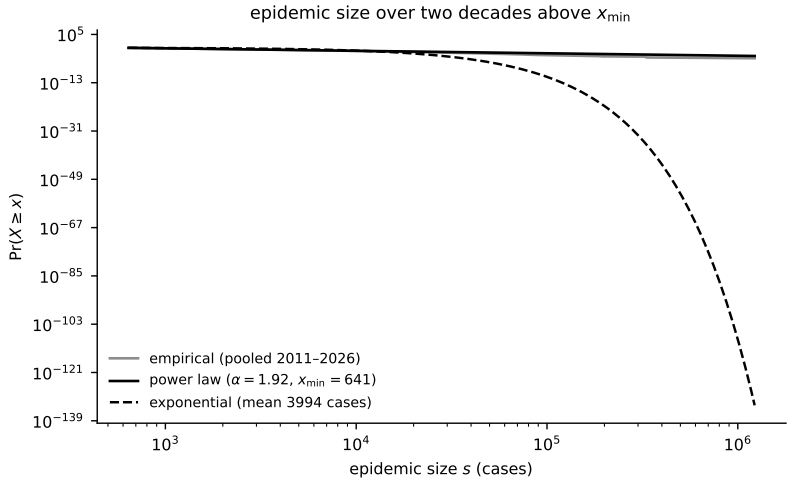


Figure 6: Empirical complementary cumulative distribution of epidemic size at the national cross-section, pooled over the 16 seasons, with the fitted power law and the exponential alternative superimposed. The empirical curve is close to linear on logarithmic axes over two decades above $x_{\min} = 730$ cases, which is the visual form of the claim used in the remainder of the paper, and the exponential alternative bends away from it over the same region.

3.3 What the exponent represents: a readout of seasonal scale

The fitted pair is not a pair of independent coefficients, and the direction of their dependence determines how the description can be used. Table 6 collects the association between the two estimates at the levels where it can be measured, together with the association recovered from simulated data in which the true exponent was held fixed.

The association is strong at national level and strong *within* states over time — positive in 92% of the 24 states with at least six seasons of data, and significantly positive in 67% — but weak between states. It is therefore a within-unit, over-time relationship and not a contrast between places. The control recovers no such association, so the estimator does not generate it: it places the sampling dispersion of $\hat{\alpha}$ at 0.03 (40 replicates, samples of 800 to 4,000 points, true exponent fixed at 2.0), against an observed standard deviation of 0.150 across the 16 national fits. The season-to-season movement in α is real, roughly five times the sampling noise in standard deviation, even though it remains small next to the variation in the case counts themselves.

Table 6: Association between the estimated exponent and the selected lower bound. The first three rows are the surveillance data; the last row is a control in which data were generated from a power law with true $\alpha = 2.0$ while the true lower bound was varied over the same range, so that any association recovered there would be induced by the estimator. Values are Spearman rank correlations between α and $\log_{10} x_{\min}$.

Stratum	Sample	ρ	n
National cross-section, season to season	16 seasons	0.89	16
Within state, season to season	median of 24 states	0.64	–
Between states	state means	0.34	26
Simulated, α fixed at 2.0	control	−0.20	40

Read as description, this is the most informative single result about $(\alpha, x_{\min})$ in the paper. Because $x_{\min}$ is measured in cases, it tracks the magnitude of the season: seasons in which municipalities produce large epidemics place the threshold higher, and the exponent moves with it. The pair is therefore a two-number summary of how large a season was, available for every municipality, far less volatile than the case count it summarises, and stable enough to be compared across seasons (Figure 7). Read as a predictor set, the same fact is a warning: an exponent that co-varies with the scale of the season is, to that extent, a re-encoding of S_{t-1}, which any model of size already carries. Both readings are correct, and they make different predictions about the ablation of Section 3.6.

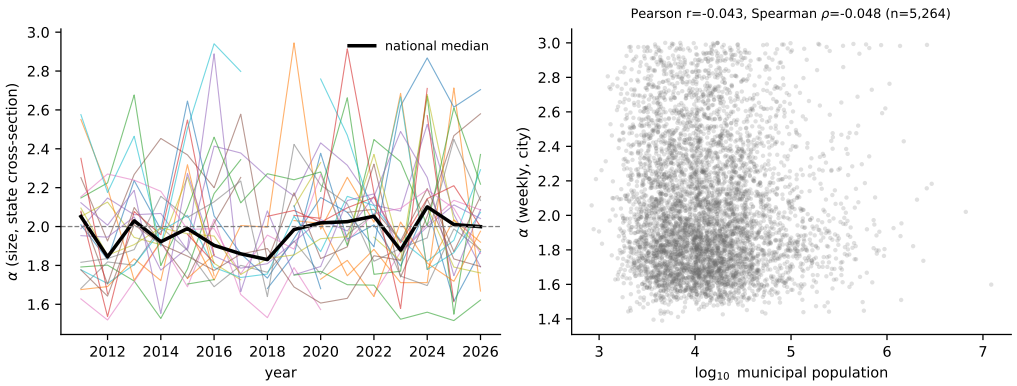


Figure 7: Stability of the exponent. (a) Estimated α for epidemic size by season, one line per state, at the state cross-section: the lines are flat, and the spread between states is far smaller than the season-to-season variation in case counts. (b) α against $\log_{10}$ municipal population, which shows no association (Pearson $r = -0.043$, Spearman -0.048, $n = 5{,}264$). Panel (a) is the empirical reason the exponent can serve as a compact descriptor of a season: it varies slowly along exactly the axis in which the observables do not.

The exponent is essentially unrelated to municipality size ($\mathrm{corr}(\alpha, \log_{10}\text{population}) = -0.043$, Spearman -0.048; $n = 5{,}264$) and only weakly related to incidence (Pearson -0.143, Spearman -0.248; $n = 4{,}663$). Larger or more heavily burdened municipalities are not in a different scaling regime: the exponent is a property of the construction, and within a unit, a property of the scale of the season.

3.4 The fitted law as a risk curve

Equation (3) makes the description operational. A power law with exponent α and lower bound $x_{\min}$ assigns $\Pr(S \geq s) = \zeta(\alpha, s)/\zeta(\alpha, x_{\min})$ for every level s, so that the fitted pair is a complete statement about the frequency of extreme seasons. It requires no fitted case count, no covariate and no training period: it is available from the current cross-section alone.

Figure 8 confronts the exceedance curve implied by each national fit with the empirical exceedance frequency of the same cross-section. The agreement holds over roughly two decades of epidemic size. Numerically, the 16 national fits imply that between 5.6% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given season, and that between 0.3% and 2.7% exceed one hundred times it. These are statements on which a surveillance service can act and which a subsequent season can falsify.

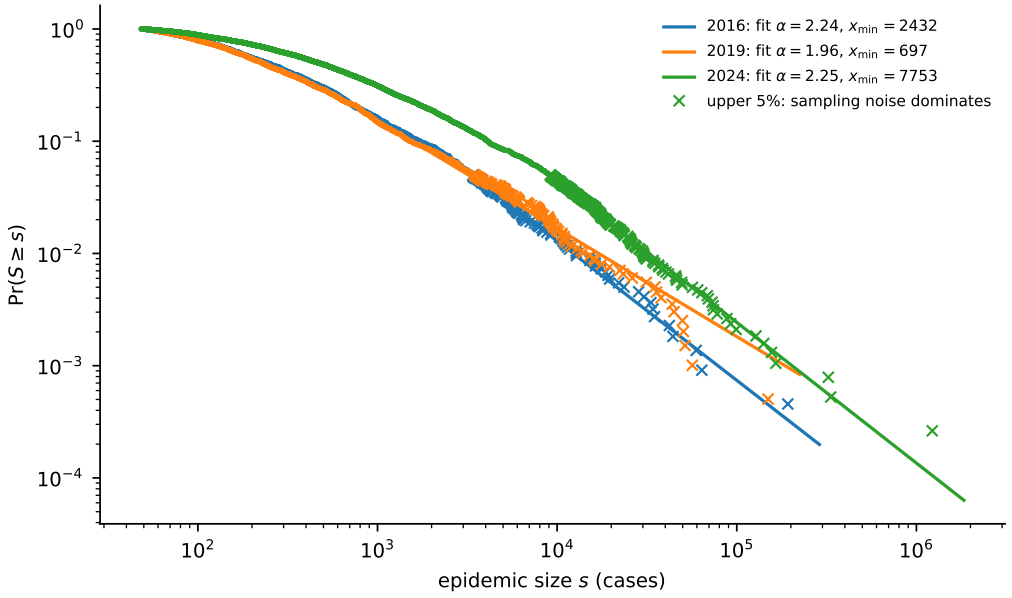


Figure 8: The fitted power law as a probabilistic statement about exceedance. For each national cross-section, curves give $\Pr(S \geq s) = \zeta(\alpha, s)/\zeta(\alpha, x_{\min})$ evaluated at the fitted $(\alpha, x_{\min})$, and points give the empirical exceedance frequency of the same cross-section. Agreement across two decades of epidemic size is what licenses the use of $(\alpha, x_{\min})$ as a compact statement about a municipality's risk of an extreme season. Where the curves separate from the points in the extreme upper tail, the sample is too small to distinguish model error from sampling noise, and those regions are marked.

This is the sense in which the description is predictive before any statistical learning is involved: α is not a summary of central tendency but a summary of how much of the risk lies in the tail, and it is available for every municipality even where that municipality has never experienced an extreme season. The familiar inference, that a divergent second moment renders extreme seasons unpredictable, is a statement about squared-error loss and not about information. Prediction of a tail event and prediction of its magnitude are distinct problems.

3.5 Predicting the probability of a large season

The exceedance formulation has a direct predictive counterpart: rather than asking how large the season will be, ask whether it will fall in the upper quartile of the unit's own history. This is the risk statement itself, and it is the natural place to look for value in the scaling descriptors, since a tail exponent is by construction a statement about upper-tail probabilities.

We posed the question at the two levels where it is asked operationally. At state level (Table 7) the lagged seasonal descriptors — which include the previous season's duration and peak timing as ordinary covariates, alongside size, R_0 and population — move the Brier score in the expected direction: a median of +0.126, positive in 4 of 5 origins and reaching +0.212. The intervals exclude zero in only 2 of the 5 origins, against the 3 that the rule requires, so the rung fails the rule at state level: each origin is a bootstrap over 23 to 25 units and the median interval is 0.26 wide, wider than the effect it is asked to resolve. Climate adds nothing to it (median +0.027) and neither does the scaling block (median +0.000, with no origin in which the block improves by an interval excluding zero). The per-origin differentials for M1 against M0 are −0.064, +0.074, +0.126, +0.212 and +0.149 for 2021 to 2025, negative in the first origin and positive thereafter, which is the pattern expected of a model that requires history in order to be trained.

At municipal level the same ladder runs over 3,976 municipalities with enough history for a within-unit threshold, at least five seasons, with between 580 and 1,484 units evaluated per origin. Two things change, and only one of them concerns the scaling descriptors. The first is a correction. With the threshold taken from each municipality's own seasons, the population-only reference falls to a median area under the curve of 0.47 (range 0.454–0.522), against 0.87 (range 0.84–0.93) where the threshold is pooled across units, as it must be at state level: a pooled label is largely a restatement of unit size, whereas the within-unit label is not. The second is that the lagged block, which the state level could not resolve, satisfies the rule here. Its Brier differential has a median of +0.049, positive and with an interval excluding zero in 4 of 5 origins, the only contrast in this study to satisfy the pre-specified rule (Table 7). Discrimination rises from 0.50 for the no-skill reference to 0.58, 0.63 and 0.65 for M1, M2 and M3; M3 is the best of the three in 3 of 5 origins and gains a median +0.013 over M2.

The inversion between the two levels is a statement about precision and not about epidemiology. The state estimate of the same effect is the larger of the two in absolute terms, +0.126 against +0.049, and the less certain by a factor of eight in interval width, 0.26 against 0.034, so that a rule which fails at one level and passes at the other is failing on sample size: twenty-five units per origin cannot resolve an effect that 1,484 units can. The two levels also differ in the scope of the label, which is pooled across states but taken within each municipality, so the state comparison inherits the size ranking described above and is the weaker of the two tests of epidemic rarity. What the two levels agree on is the ranking that matters for this paper. The scaling block satisfies the criterion at neither level: at municipal level it moves in the expected direction in 5 of 5 origins, by a median of +0.002, but its interval excludes zero in only one of them, and at state level it moves in the expected direction in 1 of 5 and has an interval excluding zero in 2, both against the block. Whether a municipality will experience an exceptional season is in part predictable from how large the previous one was; the exponent does not add to that.

One forecast origin requires comment. In 2024, 69.5% of the evaluated municipalities exceeded their own upper quartile, against a training base rate of 26.4%. The no-skill reference scores a Brier score of 0.40 there and the fitted rungs about 0.25, better than the reference but worse than the 0.21 that a predictor knowing the season's own frequency

would earn. That season was a national record, and no model trained on earlier seasons anticipated its level. This is the boundary of the exercise: the classifier ranks units within a season of familiar magnitude; it does not anticipate a change of regime.

Figures 9 and 10 report discrimination, calibration and attribution for the municipal classifier. The calibration panel is the relevant one: a classifier useful for planning must track the diagonal in the upper deciles, not merely rank well on average.

Table 7: The large-season classifier at both aggregation levels, five forecast origins 2021–2025. The target is whether a unit exceeds the 75th percentile of its own history; the loss is the Brier score. Positive values mean the later rung has the lower Brier score, and "median" is over the five origins. "Origins CI excl. 0" counts the origins with a same-direction improvement whose 95% unit-bootstrap interval excludes zero, which is what the decision rule of Section 2.11 requires in at least 3 of 5. The municipal rows use a within-unit threshold, without which the label is a population indicator; there the population-only reference reaches a median AUC of 0.47 against 0.58–0.65 for the fitted rungs.

Level	Contrast	ΔBrier, median	Origins $\Delta > 0$	Origins CI excl. 0	
State (26 units)	M1−M0 (lagged metrics)	+0.126	4/5	2/5	n
	M2−M1 (climate)	+0.027	3/5	2/5	n
	M3−M2 (scaling)	+0.000	3/5	0/5	n
Municipal (3,976 units)	M1−M0 (lagged metrics)	+0.049	4/5	4/5	
	M2−M1 (climate)	+0.003	4/5	2/5	n
	M3−M2 (scaling)	+0.002	5/5	1/5	n

3.6 Predicting epidemic size

The harder question is whether the tail helps forecast the magnitude of a season, which is the quantity against which a health secretariat plans capacity. Here the answer is negative, and the reason is the coupling measured in Section 3.3.

Table 8 reports the ladder over the five origins at state level. Against calibrated persistence, the lagged-descriptor rung M1 carries essentially no skill for size (median skill 0.30, range −0.06 to 0.86, positive in 3 of 5 origins and with an interval excluding zero in 2 of 5), and successive rungs do not improve on it: M2 has median 0.12 and M3 0.15. The differential of M3 against M2, the one that isolates $(\alpha_{t-1}, x_{\min,t-1}, \dot{\alpha}_{t-1})$ and nothing else, shows no origin in which the block improves with an interval excluding zero, against the three the rule requires; its interval excludes zero in 2 of the 5 origins, both against the block, with a median relative change in pinball loss at $\tau = 0.5$ of −5.2%.

The block was also expected to be most valuable at high τ. The argument for α_{t-1} is that it encodes how much of the risk lies in the tail, so that any gain should concentrate at high τ, and Figure 12b was constructed to test precisely that. It shows not a degradation but an absence of signal. At $\tau = 0.95$ the median differential is +4.9%, with M3 ahead, and it is positive in 4 of 5 origins; but the intervals exclude zero in only 2 of 5, and those two point in *opposite* directions, +32.7% in 2021 and −15.4% in 2024. The same holds across the grid: the median of the per-origin differential moves between −13.6% and +7.2% with no trend, and there is no quantile at which 3 of 5 origins agree in direction with an interval excluding zero, 0 of 19 quantiles clearing the criterion. This is what the coupling predicts. Both statistics are functions of the magnitude of the previous season,

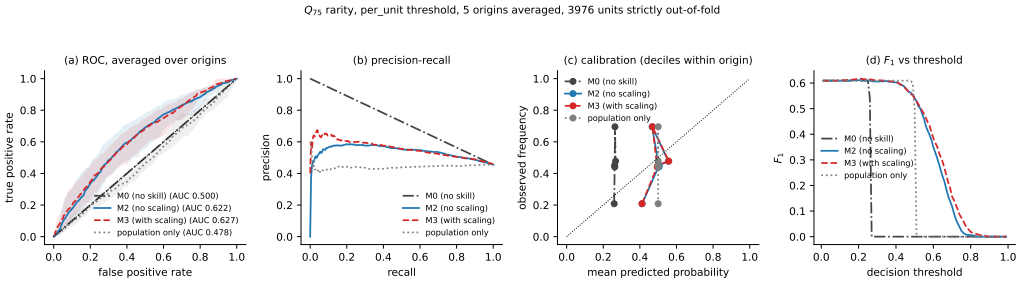


Figure 9: Classification of large seasons at municipal level. (a) ROC curves; (b) precision–recall curves, the more informative summary when the positive class is the minority; (c) calibration of the predicted probability of exceeding the 75th percentile; (d) F_1 as a function of the decision threshold, with the Youden point marked. All curves are computed out of fold and averaged over the five forecast origins rather than pooled across them: pooling is not neutral here, because the base rate ranges from 0.21 to 0.70 between origins and a prediction that knows only the calendar season reaches a pooled area under the curve of 0.60 while having no within-origin discrimination at all. Solid lines denote M2, without scaling descriptors, dashed lines M3, with them, the dash-dot curve the no-skill M0 reference and the dotted grey curve the population-only reference.

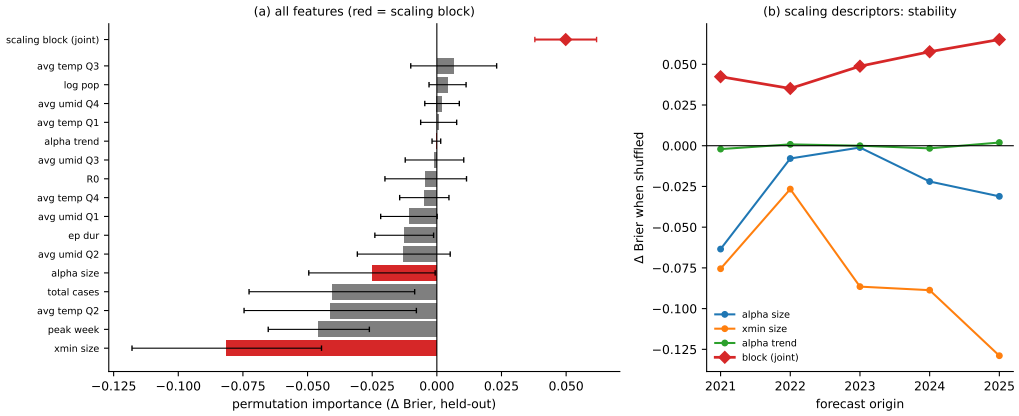


Figure 10: What the classifier uses. Held-out permutation importance, measured as the increase in the Brier score when a predictor is permuted, so that a negative value means the predictor is detrimental. Panel (a) ranks the individual predictors, with the power-law block in red; panel (b) tracks the three scaling descriptors across forecast origins. The joint bar is the scaling block permuted as a single unit: permuting one column while its correlated partners remain in place understates a collinear block, and α_{t-1} and $x_{\min,t-1}$ are rank-correlated at 0.64 within units.

Table 8: Ablation ladder over the five forecast origins 2021–2025, state level (26 units, 23–25 evaluated per origin), for the single target. Cells give the median across origins of the skill score $1 - L(\text{rung})/L(\text{M0})$ against calibrated persistence, with the range over origins in brackets; the primary loss is the pinball loss at $\tau = 0.5$. The last column is the M3−M2 differential at that loss: median relative change, positive meaning that the scaling block improves the forecast, and in parentheses the number of origins in which M3 does so by sign alone. The classifier is reported separately in Table 7.

Target	M1	M2	M3	M3−M2
Size S_t	0.30 [−0.06, 0.86]	0.12 [−0.39, 0.86]	0.15 [−0.77, 0.81]	−5.2% (1/5)

so that the block is at best a second and noisier copy of S_{t-1}; the ablation shows that the copy carries no usable increment, which leaves the upper quantile where it was rather than displacing it in either direction.

Two origins dominate the size target and require separate treatment. The 2024 season was the largest on record and 2025 fell back sharply; persistence extrapolated the record into 2025 and, in absolute terms, lost there much of what it had gained in 2024. The pinball losses for size are accordingly an order of magnitude larger in those two origins, 1.5 to 1.7×10^5 cases, than in the other three, 1.2 to 3.2×10^4, so that a pooled mean over origins would mostly describe how each rung behaves across a change of regime. We therefore report the median across origins and the sign of the per-origin differential rather than a pooled average.

That change of regime is the substantive lesson of this subsection, and it is consistent with the descriptive result rather than in tension with it. If epidemic size follows $\alpha \approx 2$, the largest season a municipality will experience lies systematically outside the support of its observed history. A model that already knows the size of the previous season is therefore approximately correct in ordinary seasons and wrong by a large factor in the exceptional one, which is exactly the pattern of the table and of Figure 11. The choice of estimator stated in Section 2.9 determines how that pattern is to be read, and not the pattern itself. Because the estimator can produce a value beyond its training responses, a large error in a record season is a statement about the data, and the failure of the size forecast can be read as a finding rather than as a property of the fit.

3.7 Error structure and spatial output

Accuracy pooled over units is not the same as accuracy where decisions are taken, so we report the error structure (Figure 13) and the map a surveillance service would read (Figure 14).

4 Discussion

4.1 What the exponent is, and what it is not

The comparison across five constructions settles a question that the applied literature has treated as empirical when it is largely definitional. Median exponents span 1.95 to 2.08 across constructions of the same series, the share of fits favouring a power law ranges from 23% to 100%, and a single change of temporal window moves support from 77% to 23% with no transmission parameter altered. What does not survive is the inference from

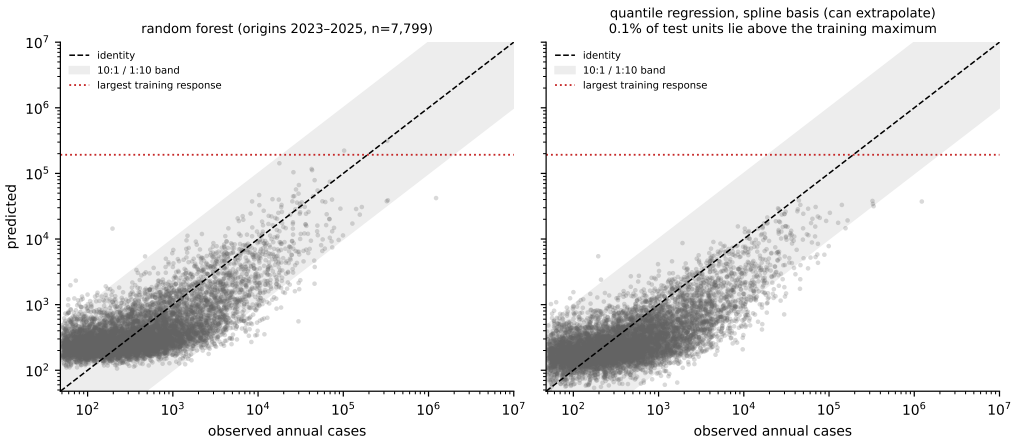


Figure 11: Out-of-sample forecasts of epidemic size, predicted against observed annual cases on logarithmic axes, one point per unit and forecast origin. Left, a random forest; right, the spline-basis quantile regression described in Section 2.9. The identity line is dashed, the 10:1 and 1:10 bands are shaded, and the dotted red line marks the largest annual total in the training window, so that points lying above it are seasons larger than any on which the model was trained. Because $\alpha \approx 2$, a handful of points in the upper right dominate any squared-error summary, which is why the primary loss in Table 8 is the pinball loss rather than the root mean squared error.

an exponent to a regime: the step that treats α as a property of the pathogen or of the place, when it is a property of the tuple (variable, sampling unit, dimension of pooling, comparison model).

The same comparison identifies which construction *does* carry a stable tail. Epidemic size across units within a season is power-law at every construction at which $x_{\min}$ lies above the bulk: all 16 national cross-sections, 96% support across 353 state-year cross-sections, and a lower bound in the hundreds of cases, two to three orders of magnitude above the body of the distribution. That is not a fragile result, and it is the one on which the paper builds.

The distinction between this result and the weekly-count fits that collapse to $x_{\min} \leq 5$ is substantive rather than technical. At $x_{\min} = 2$ the fitted law describes counts of one and two cases; at $x_{\min} = 730$ it describes the upper decade of a distribution. Two analyses reported as agreeing on the exponent can be describing entirely different objects, as the comparison of W-mun and S-mun shows: a difference of 0.13 in the exponent and a factor of 150 in the lower bound.

4.2 The tail as a risk specification rather than a forecast input

The most consequential move in this paper is to stop treating $\alpha \approx 2$ as a verdict on forecastability and to treat it as a probability statement. A power law assigns $\Pr(S \geq s)$ at every level; what it precludes is that the *magnitude* of the largest event be estimated from the variance. These are distinct claims, and conflating them is what produced the intuition that heavy-tailed epidemics are unforecastable.

Read as a risk specification, the fitted pair performs well and cheaply. Two parameters reproduce the empirical exceedance frequency over two decades of epidemic size

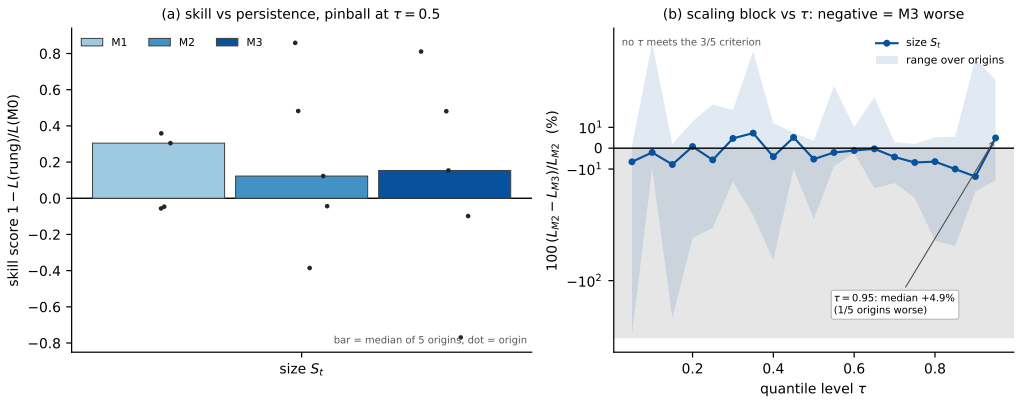


Figure 12: The ablation ladder for the size target. (a) Skill score relative to the calibrated persistence benchmark (M0) at the primary loss, the pinball loss at $\tau = 0.5$; bar height is the median over the five forecast origins and each dot is a single origin. The step from M2 to M3 isolates the contribution of the scaling descriptors. (b) The quantity on which the decision rule turns, $100\,(L_{M2} - L_{M3})/L_{M2}$, against τ: a gain confined to high τ would indicate that the scaling information concerns extreme seasons specifically, which is what a heavy-tail description should deliver, so that negative values here are the opposite of the intended effect. Lines are medians over origins and the band is the range across origins. At $\tau = 0.95$ the median differential is $+4.9\%$ and positive in 4 of 5 origins, so that the panel shows an absence of signal rather than a gain: the intervals exclude zero in only 2 of 5 and point in opposite directions. Skill scores carry no interval of their own, the unit-bootstrap interval belonging to the paired differential.

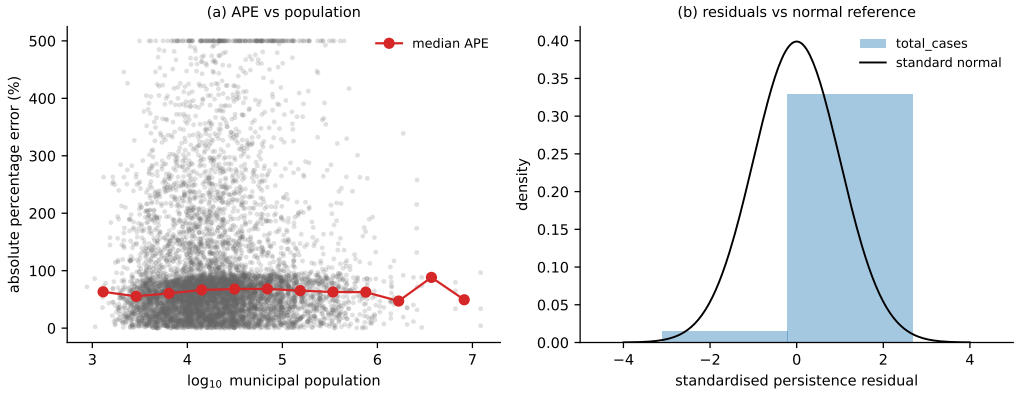


Figure 13: Forecast error against population size. Absolute percentage error of the size forecast by unit population, one point per unit and origin, with a locally weighted smooth. Errors are largest for the smallest municipalities, where case counts are low and a single imported case represents a large relative change, and they narrow but do not vanish at the top of the size distribution.

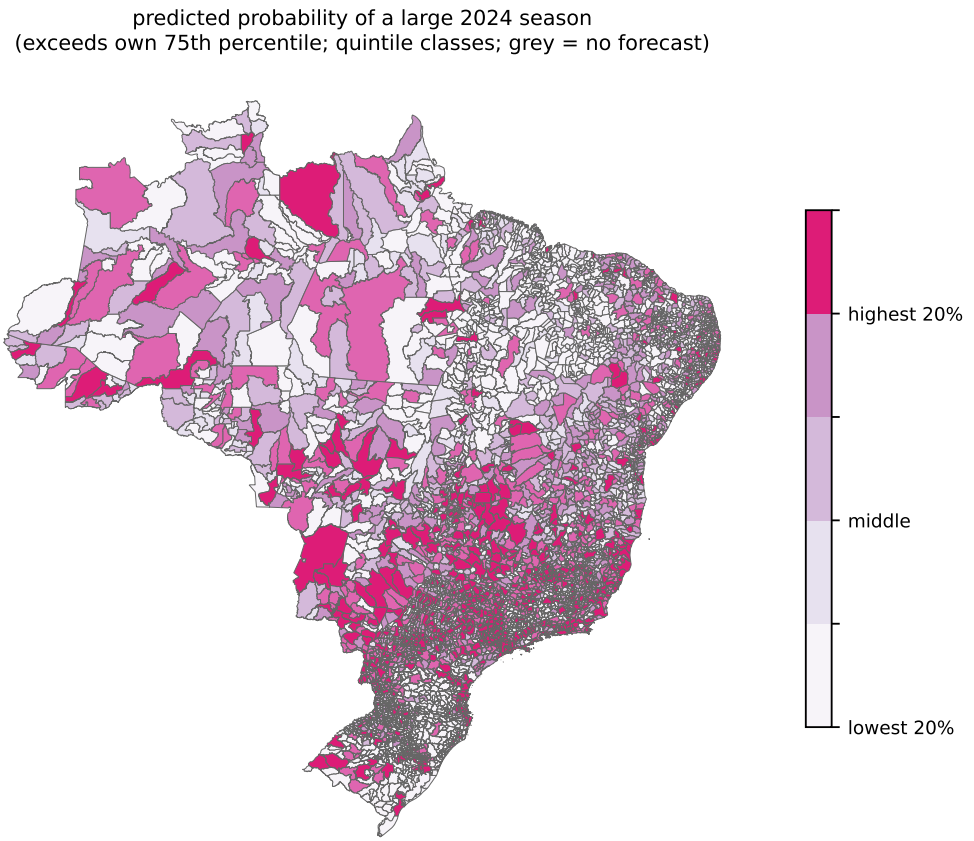


Figure 14: Out-of-sample forecasts of epidemic size by state for one forecast origin, in the form in which a surveillance service reads them. The map shows the median quantile forecast of annual cases; the legend is truncated at the 95th percentile of the plotted values so that the largest states do not flatten the colour scale for the remainder.

(Figure 8), and they imply that between 5.6% and 16.5% of municipalities exceed ten times $x_{\min}$ in a given season against 0.3–2.7% exceeding a hundred times it. No covariate, no training period and no fitted case count is involved: the statement is available from the current cross-section, for every municipality, including those that have never experienced an extreme season. For a service deciding how much capacity to hold in reserve, this is closer to the decision than a point forecast is.

The predictive evaluation confirms the division of labour and locates it more sharply than a single level could. Where the question is posed as a probability — will this municipality exceed its own upper quartile? — the answer is affirmative, but only where the panel can resolve it. At municipal level lagged seasonal descriptors improve the Brier score over calibrated persistence in 4 of 5 origins and are significantly the worse in the fifth, and they raise discrimination from 0.50 to 0.58 in area under the curve, the full ladder reaching 0.65. At state level the same rung produces a larger point estimate and fails the rule, on 23 to 25 units per origin and with a label pooled across them. The scaling block, which is the object of the paper, satisfies the criterion at neither level. Where the question is posed as a magnitude, the answer is negative, and the failure is orderly rather than unexplained.

4.3 How the coupling explains both results

The coupling between α and $x_{\min}$ is the mechanism behind both halves of the predictive result, and it merits reading as a finding rather than as a caveat. Within units the two estimates are rank-correlated at 0.64 (median over 24 states) and at 0.89 nationally, whereas a simulation with the true exponent held fixed at 2.0 recovers -0.20. The association is a property of the data and not of the estimator.

Descriptively, this makes the pair a compact readout of seasonal magnitude: two numbers that move with the size of the season, vary far less than the case counts they summarise (SD 0.150 against a sampling noise of 0.03), and are available for every unit. This is a substantive contribution to the characterisation of a season, since it reduces a quantity spanning two orders of magnitude to a stable low-dimensional summary.

Predictively, the same fact accounts for the null. An exponent that co-varies with the scale of the season is, to that extent, a re-encoding of S_{t-1}, which the model already contains. The ablation measures that redundancy directly: the scaling block adds nothing to a rung that already knows the previous season's size, in 0 of 5 origins, and at $\tau = 0.95$ its differential is $+4.9\%$ in median with intervals that exclude zero in only 2 of 5 origins and in opposite directions, so that the design has no power there to sign the effect in either direction. The redundancy was predictable from the coupling before the ablation was run, and the ablation confirmed it where the prediction had content: a re-encoding of a variable already in the model must deliver no incremental information, and none appears at any quantile of the grid.

The same reasoning explains why the exceedance question behaves differently. Asking whether a season will be large is a question about upper-tail probability, and the coefficient on the lagged descriptors is doing the work there; adding a transformation of the same lagged quantity cannot improve it, which is why the Brier differentials of the scaling block sit near zero with no interval excluding it. The information the exponent carries is real, and it is already contained in the lagged count.

4.4 Size as the hard target, consistently with the description

Nothing in this paper shows that epidemic size can be forecast well. Against calibrated persistence no rung achieves skill in more than 3 of 5 origins, and 2 of the 5 origins dominate the loss because they straddle a change of regime: 2024 was the largest season on record and 2025 fell back sharply.

The failure is not an artefact of the estimator. The primary model is a spline-basis quantile regression, which by construction continues the boundary trend beyond the largest response on which it was trained (Figure 11), so that a record season is a statement about the data rather than a property of the estimator. The quantile regression at $\tau = 0.95$ represents the upper region instead of interpolating within it, and the forecast still fails in 2024: knowledge of the previous season's size and duration, of its peak week, of the climate covariates and of the scaling descriptors does not order five origins at 23 to 25 units each.

The defensible summary is therefore a division of labour rather than a verdict. The tail does not indicate how large the next season will be; it indicates how likely an extreme season is, and does so with two parameters and no training data. These are different products for different decisions, and the second is the one a surveillance service can use before the season begins.

4.5 A reporting convention

The practical consequence of the descriptive half is a reporting convention, and it is inexpensive to adopt. An exponent should travel with the lower bound that produced it, the construction of the sample, the number of observations and the alternative model against which it was tested. Of these, $x_{\min}$ is the most informative and the most frequently omitted, because it separates two analyses that look identical in a table of results. The diagnostic is easy to apply and difficult to manipulate, since $x_{\min}$ is selected by the same goodness-of-fit criterion that selects the model and cannot be tuned to produce a desired α. We would argue further that an α reported without $x_{\min}$ should be treated as an incomplete statistic, in the same way that a correlation is incomplete without its sample size.

We would add a second convention, specific to the argument tested here. A claim that a scaling description improves forecasts is meaningful only against a specified benchmark, on a proper scoring rule, out of sample, with the scaling block isolated from the blocks to which it is added, and with every channel of leakage closed. The redundancy documented here would be invisible under a pooled coefficient of determination without a benchmark, since between-unit differences are reproduced trivially by persistence.

4.6 Relation to operational forecast systems

Operational arbovirus platforms — InfoDengue and Mosqlimate in Brazil, and the national systems reviewed elsewhere — already produce weekly municipal nowcasts and short-horizon forecasts (Araujo et al., 2025). Alongside them, multivariate machine-learning models have been fitted to municipal dengue series in Brazil, comparing recurrent networks with random forest regression (Mussumeci and Coelho, 2020). The operational systems answer the question that this paper finds difficult: what is happening now, and what will happen in the coming weeks. The scaling description answers a question that they largely do not: given the history of a municipality, how often should its cross-section produce a season of a given size, expressed as a probability with two parameters. The

two are complementary, and the second is inexpensive enough to be computed for every municipality in the country from a single cross-section.

4.7 Limitations

Several limitations bound these conclusions.

The scaling block does not improve the size forecast. This is reported in the Results rather than only here because it is a measured outcome, but it is the principal negative result of the paper and should accompany any use of it. Against a benchmark that already carries the lagged case count, the block of $(\alpha_{t-1}, x_{\min,t-1}, \dot{\alpha}_{t-1})$ fails the pre-specified test on size in 0 of 5 origins. At the 95th percentile its differential is +4.9% in median, with intervals excluding zero in only 2 of 5 origins and in opposite directions, so that the design cannot sign the effect there: with 23 to 25 units per origin the test is underpowered at the extreme quantile, and we report the absence of a measured gain rather than a measured deterioration. The mechanism is the coupling documented in Section 3.3. A rung containing the scaling descriptors *alone*, without the lagged descriptors, would answer a different and interesting question — whether three power-law parameters can substitute for the full set of covariates — and is not reported here.

The municipal classifier is a ranking device, not a level forecast. With the exceedance label referred to each municipality's own history, the population-only reference falls to a median area under the curve of 0.47, so that the task no longer measures unit size, and the lagged block satisfies the pre-specified incremental test at that level. What bounds the claim is the purpose of the classifier: discrimination reaches a median of 0.65 only with the full ladder, against 0.58 for the rung that satisfies the rule, the scaling block adds a median +0.002 in Brier score, and the 2024 origin, in which 69.5% of units exceeded their own upper quartile against a training base rate of 26.4%, defeats every rung. The classifier orders municipalities within a season of familiar magnitude. It does not anticipate a change of regime, and in a record season its probabilities should not be read as levels.

The case series are nowcast-adjusted. The analysed variable is the estimated case count produced by the nowcasting model of the platform, not the raw notification count. This is the appropriate series for a retrospective description, because raw notifications are incomplete in recent weeks, but the upper tail inherits whatever smoothing the nowcast applies. If the nowcast shrinks extreme values, the fitted exponents are biased towards the bulk, and the size distribution reported here is somewhat less heavy than the notification distribution would be.

The state level is an arithmetic aggregate. Where duration and peak timing enter as covariates, the state values are the mean of municipal durations and the median of municipal peaks. Neither is the duration or the peak of the aggregate epidemic, which is the quantity a state health secretariat actually experiences. The state-level results should be read as properties of the distribution of municipalities within a state, not as properties of the epidemic of the state.

The cross-sections are short. The national constructions rest on 16 annual fits and the state constructions on three to 16 seasons per state, so that the correlations reported in Section 3.3 are estimated on small samples even where the underlying cross-sections contain thousands of municipalities. The within-state estimates are more robust than the national ones because they pool over 24 states.

Extrapolation rests on an assumption rather than on data. The primary estimator continues the boundary trend of the spline basis linearly, which is what allows it to name a season larger than any in its training window; but the value it produces in that region is an extrapolation of a local slope, and it is only as reliable as the assumption that the

slope holds. In a simulation constructed at the scale of this panel, with a season twice the previous one and training seasons of 10 to 1,000 cases, the model recovered the doubling to within 0.1% at five times the largest value in its training window and within 0.2% at twenty times. This establishes that the estimator recovers the relationship when the relationship is of the assumed form; it does not establish that the relationship holds for dengue. The defensible reading is that the estimator removes the ceiling on the magnitude the forecast can take, so that a large error in a record season can be attributed to the data, and that it does not remove the risk of a large error.

The classification threshold is a choice. Exceeding the 75th percentile of a unit's own history defines a large season; at the 90th percentile the positive class becomes small enough that the balanced accuracy of any classifier falls and the comparison between rungs loses power. The threshold is therefore a design parameter, fixed before the evaluation and reported with it.

The exponent cannot exceed three. The estimator bounds α above by 3, so that a steeper tail is reported as 3 rather than as its unconstrained maximum. This is immaterial here: no national or state-year size fit approaches the bound, the largest values being 2.94 at state level and 2.25 nationally, and it is reached only by 12 of the 48,937 weekly fits, or 0.02%. Every exponent reported in this paper is an unconstrained maximum.

Predictors are ecological and coarse. No serotype composition, no seroprevalence and no vector index enters the model, although all three are plausible determinants of season size. None of the three is observable with the data at hand: the attack ratio can be recovered from aggregated incidence only under strong assumptions about the initial susceptible fraction, and serological surveys are what would be needed to characterise serotype-specific immunity (Coelho and Carvalho, 2015); where susceptibility has been built into municipal dengue forecasts it has improved their weather-based predictions, which is reason to expect its omission to matter here (McGough et al., 2021). To the extent that these are stable within a unit they are absorbed by unit-level effects; to the extent that they vary by season they constitute unmodelled heterogeneity that the scaling block may be inadvertently representing. This is a tenable explanation for the null reported here, and it is the first that we would test.

4.8 Implications for surveillance practice

Three practices follow, none of them costly.

First, report the tuple. An exponent accompanied by its construction, its $x_{\min}$, its sample size and its comparison model can be assessed by a reader; an exponent alone cannot, and the five constructions examined here show that the same exponent can carry opposite meanings.

Second, use the exceedance curve. Two parameters fitted to the current cross-section give the probability that a municipality will exceed any level, and the national fits reproduce the observed frequency of large seasons over two decades. This is an inexpensive input to planning that requires no model training and no history beyond the current season.

Third, do not translate a heavy tail into a claim of unpredictability without specifying the loss function. A distribution without a finite variance constrains squared-error point forecasts and leaves probabilistic exceedance statements intact, and the exceedance statement is usually the one a health service requires.

5 Conclusion

The distribution of dengue epidemic size in Brazil is heavy-tailed with an exponent close to two, and this is among the few robust facts in the applied literature on epidemic scaling: 16 of 16 national cross-sections, 96% of 353 state-year cross-sections, and a lower bound two to three orders of magnitude above the bulk. Which of these statements a study reports, and whether it reports a power law or an exponential at all, is determined by the construction it imposes on the data; that heterogeneity is methodological before it is biological.

The consequence for forecasting is not the one usually drawn. Because $\alpha \approx 2$ fixes the entire exceedance curve, the fitted pair is a two-parameter risk specification, and it is well calibrated against the empirical frequency of large seasons over two decades of epidemic size. Posed as a probability, the question of whether a municipality will exceed its own upper quartile is tractable: lagged seasonal descriptors improve the Brier score over calibrated persistence in 4 of 5 municipal origins, the only contrast here to satisfy the pre-specified rule, and raise discrimination from 0.50 to 0.58 in area under the curve, the full ladder reaching 0.65, whereas at state level the same contrast cannot be resolved on 25 units. Posed as a magnitude, it is not tractable: no rung achieves skill for size, and the scaling block specifically adds nothing, in 0 of 5 origins, its differential at the 95th percentile being +4.9% in median with intervals that exclude zero in only 2 of 5 origins.

That last result is not an accident of the data. The exponent co-varies with the lower bound, and hence with the scale of the season, at 0.89 nationally and 0.64 within states, in a manner that simulation does not reproduce. The pair is therefore a genuine low-variance summary of how large a season was, and for that same reason a redundant input to any model that already knows how large the previous one was. The redundancy was predicted from the coupling before the ablation was run, and the ablation confirmed it.

What generalises beyond this dataset is the shape of the question. A heavy tail does not render epidemics unpredictable; it bounds how well the magnitude of the largest event can be estimated from second moments, and it leaves the probability of exceeding any level fully specified. The exponent is worth reporting. It is worth reporting as a risk curve, with its lower bound, and not as a predictor.

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